

Project Proposal (OOP)

Title: Library Book Management System

Problem Statement

The proposed Library Book Management System will allow librarians to track the returned and issued books, and maintain records of library members. The system will provide functionalities such as adding new books, issuing books to users, returning books by user, and searching for books. This project will demonstrate the use of Object-Oriented Programming concepts in Java to design a modular, maintainable, and efficient system while ensuring proper handling of errors such as issuing unavailable books or invalid member details.

Target User

Library staff managing books, records of members in a library in college, school or govt libraries.

Core Features

- Addition of new books
- New library members add
- Issue books to members
- Return the books taken by members in library
- Books can be searched by their id
- Display the books which are issued and available

OOP Concepts to be Used

1. **Abstraction:** Abstract classes define general properties and behaviors of library entities such as books and users.
2. **Inheritance:** Different types of users such as Student Member and Faculty Member extend from a base Member class.
3. **Polymorphism:** Methods such as issue of books or fine calculation work differently for different members.
4. **Exception Handling:** It Handles invalid inputs such as books not returned or unavailable books.

Proposed Architecture Description

The system uses an object-oriented design with classes such as Book, Student member, faculty member and a librarian. The Book class stores book details such as book id, title etc. Member represents users who borrow books. A librarian controls issue of books, adding of new books and arranging them.